

CII SR -14th Edition Kaizen Compétition 2019

A Kaizen Presentation By



EVEREST INDUSTRIES LIMITED
(Podanur Works – Coimbatore)

Company Profile

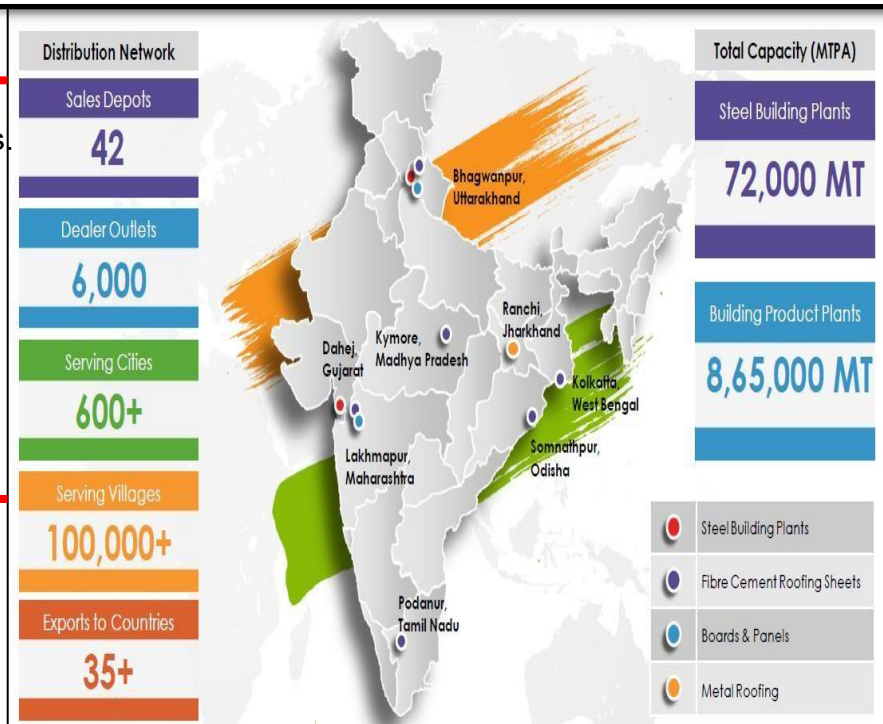


Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%)** – include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Our Esteemed Customers



Kaizen Theme



Kaizen Theme: To reduce Breakdowns of Hog Beater Gear Box.

Implemented Area: Sheeting Machine-1 Shop Floor, Podanur Works, Coimbatore

Implementation Date: September 2018

Cross Functional Team:



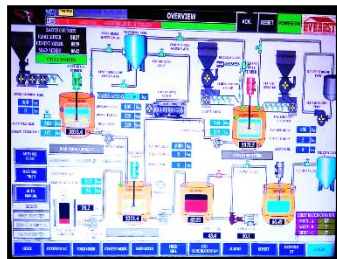
**Mr. Prasadhan
(Fitter)**



**Mr. Edward
(Fitter)**

Problem / Present Status

Process Flow:



**RAW MATERIAL
PREPARATION**



VATS



SHEET FORMING

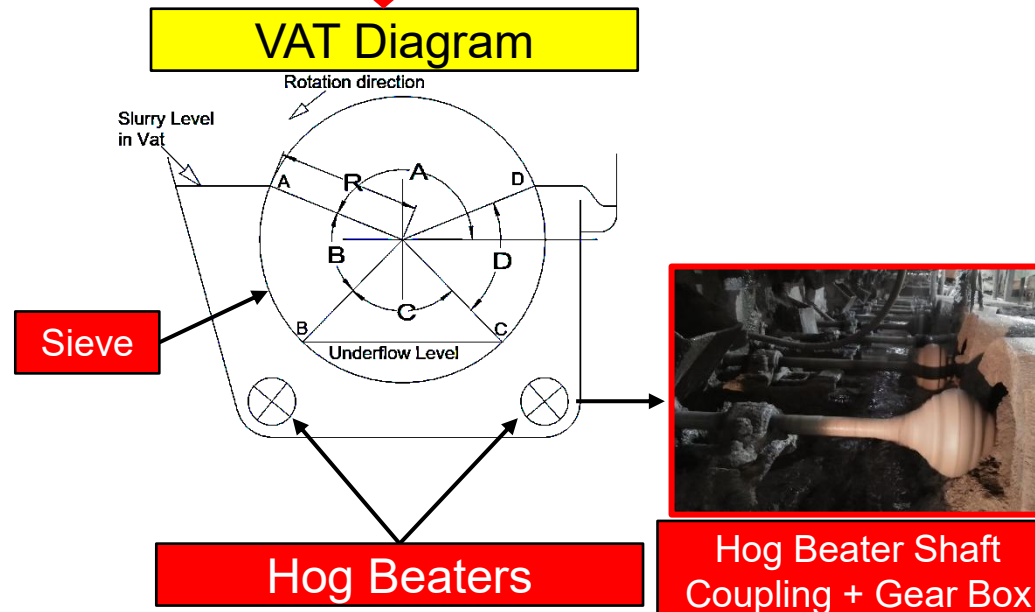


**SHEET
HANDLING**

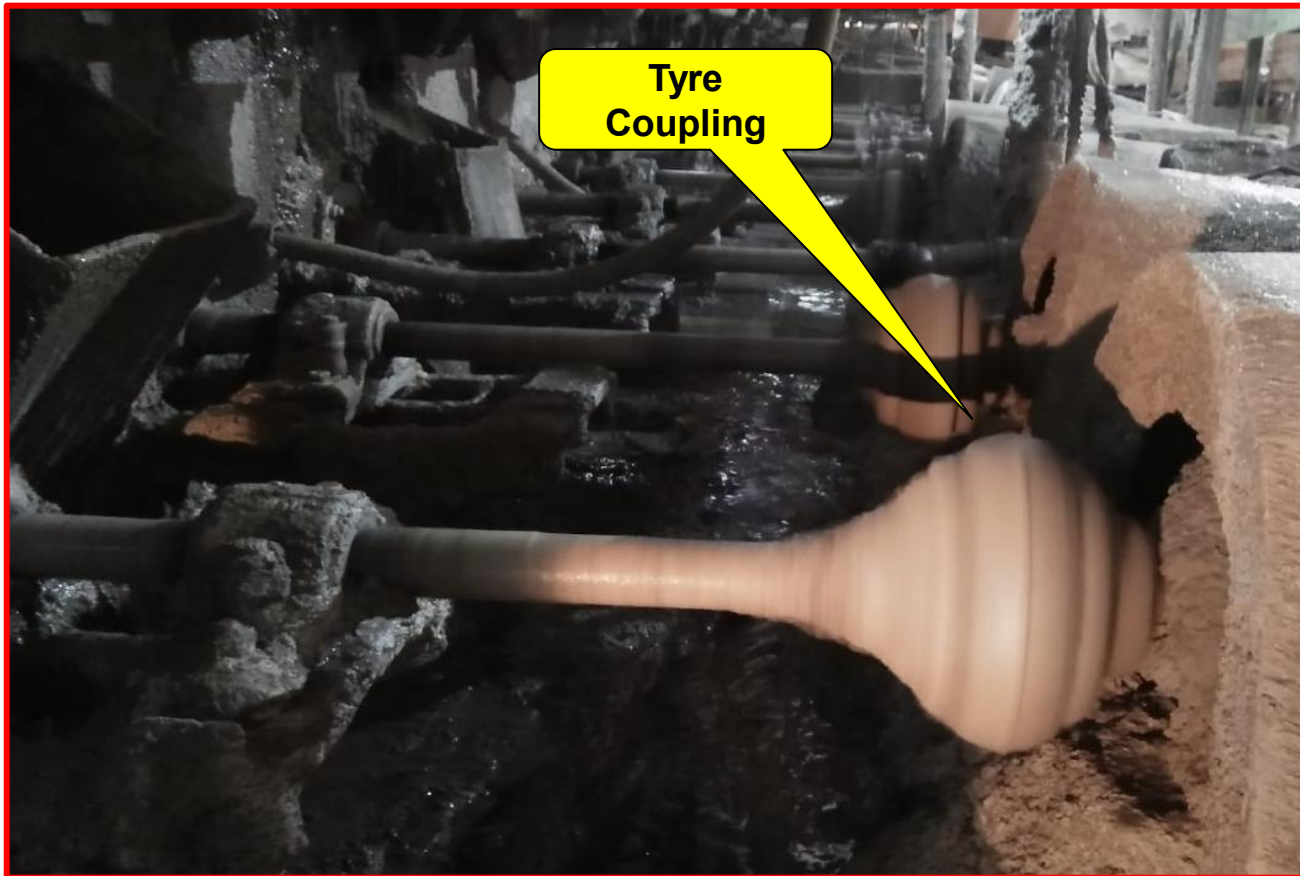


PRE-CURING

DE-PIILING



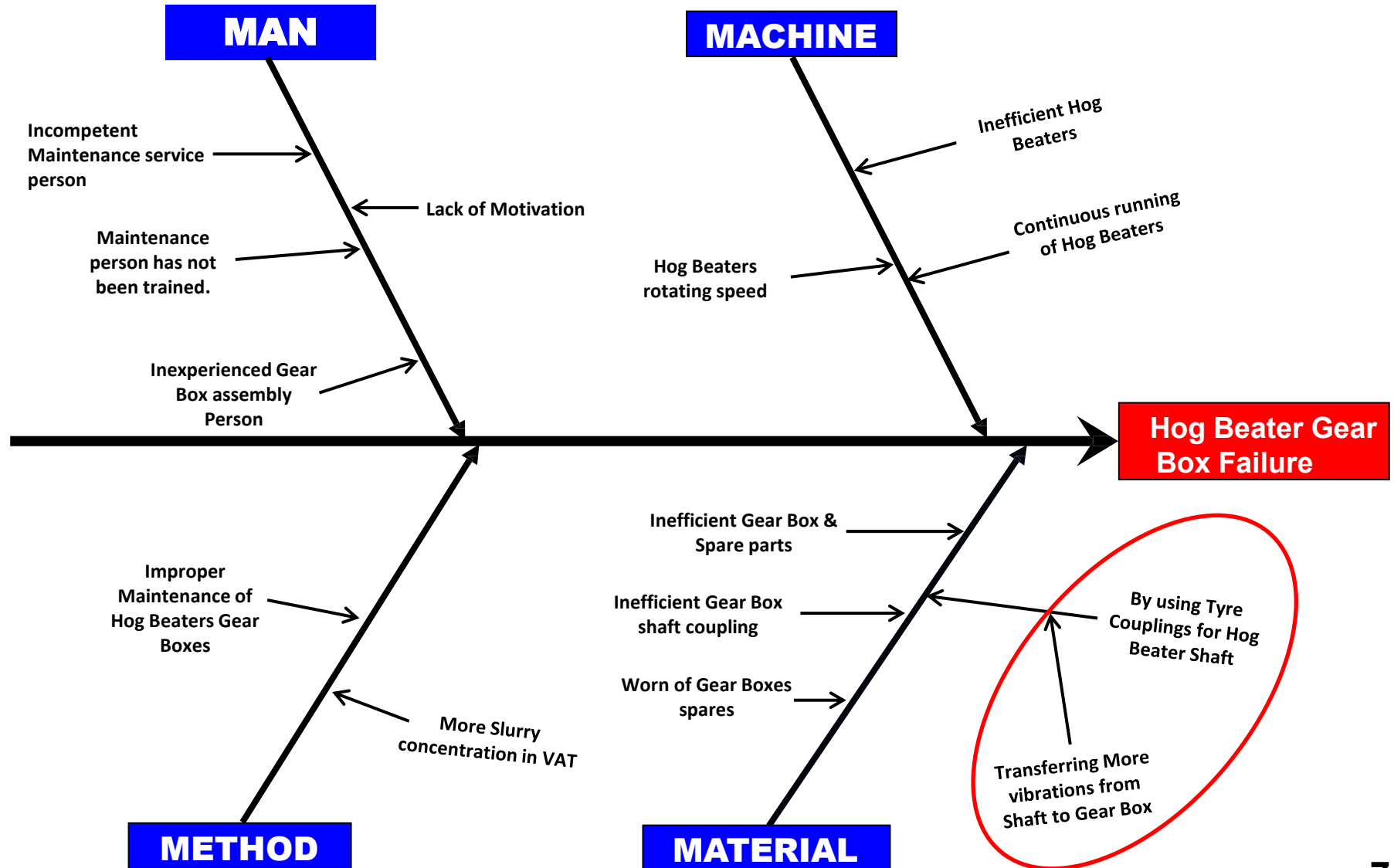
Problem / Present Status



Problem/ Present Status :

By using Tyre coupling, vibrations from Hog Beater shaft is transferring directly to Gear Box. Due to this there is frequent failure of Gear Box.

Cause & Effect Analysis



Root Cause Analysis & Idea Generation



Problem: Frequent Hog Beater Gear Box failure.

Why 1 : Gear Box is experiencing vibrations.

Why 2 : Hog beater shaft is transferring vibrations to Gear Box.

Why 3 : Hog beater has Tyre coupling and it is not absorbing vibrations and oscillations of shaft.

Root Cause: Tyre coupling is not absorbing vibrations and oscillations of the shaft.

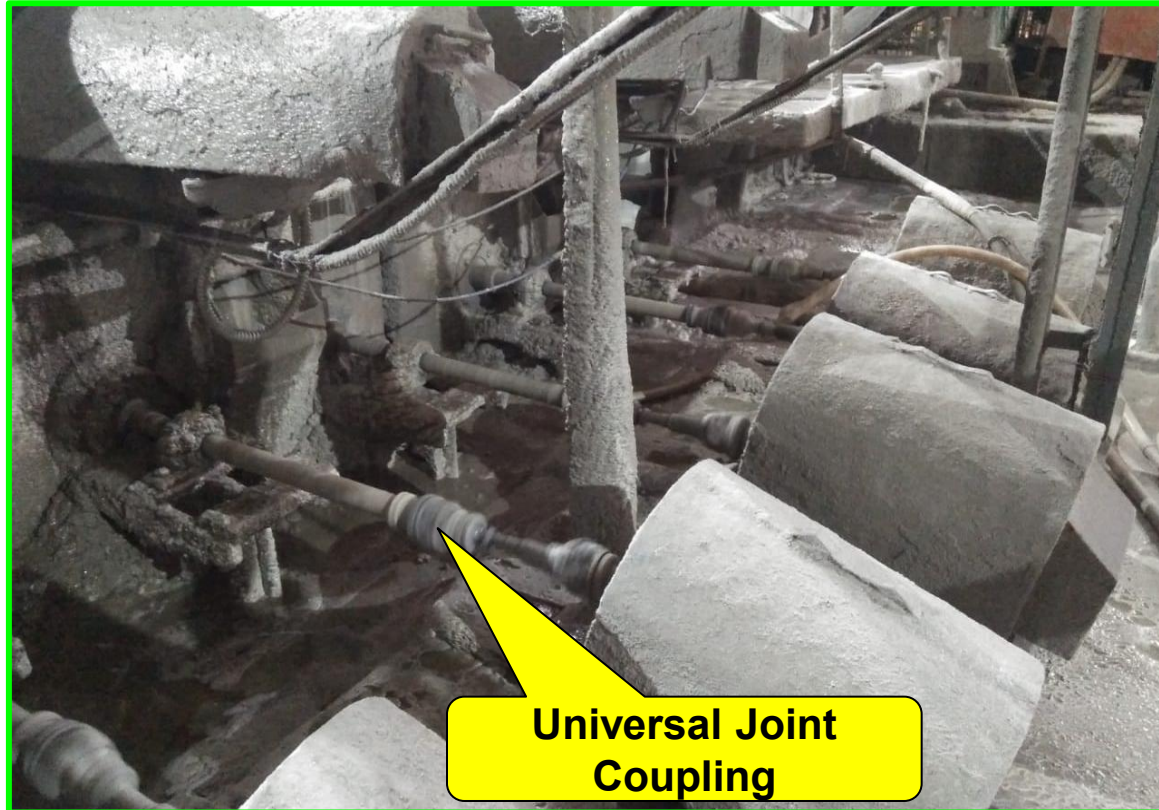
Idea Generation: To change all the Tyre couplings of Hog Beaters to Universal Joint Coupling.

Action: Replaced all the Tyre coupling of Hog Beater with Universal Joint coupling.

Countermeasure

SN	Counter Measure Activities	Target Date	Responsibility	Status
01	To observe and analyze Reasons for Hog Beater Gear Box failure.	15/08/2018	Maintenance & Production Department	Completed
02	To study about Hog Beater shaft couplings	20/08/2018	Maintenance Department	Completed
03	To change all the Tyre couplings of Hog Beater to Universal Joint coupling.	12/09/2019	Maintenance Department	Completed

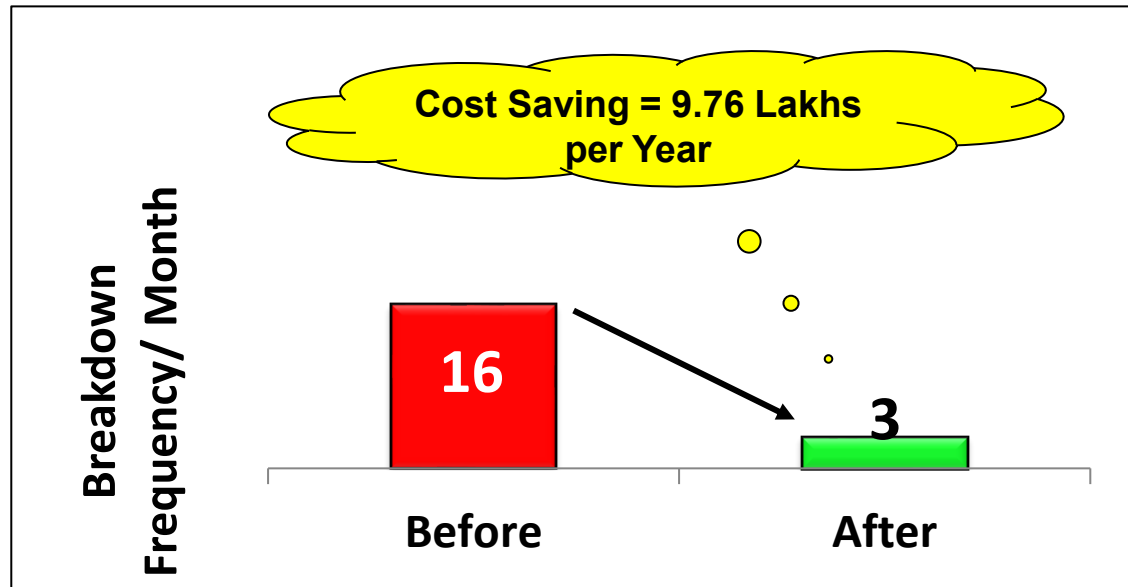
After Situation



- Replaced all the Tyre couplings of Hog Beaters with Universal Joint Coupling, which allows flexibility of the shaft while rotating. By this Gear Box failure frequency has reduced.

Validation of Data: Calculation

- Cost of 1 Gear Box servicing = 10,091 Rs/Gear Box
- Maintenance Cost of Hog Beater Gear Boxes = 83,956 Rs/Month
- Maintenance Cost of Hog Beater Gear Boxes = 10.07 Lakh/Annum



Investment Cost: Rs. 31,800/- For Universal joint Couplings

Tangible Benefits

- Out Hog Beater Gear Box failure has reduced by 81% from 16 Times per Month to 3 Times per month.
- Cost Saving of 9.76 Lakh/Annum.

In-tangible Benefits

- Morale of People increased due to reduction in frequent Breakdowns and reduction in maintenance effort of Hog Beater Gear Boxes.

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

BW	CW	KW	LW	PW1	PW2	SW

■ - Not Applicable ■ - Applicable & Deployed

Kaizen Sheet

Kaizen Sheet							Company		MM/YY	S.No												
Productivity	Quality	Delivery	Safety	Material	Energy	Maintenance	Everest Industries Ltd	09/18	3													
Kaizen Title: To reduce Breakdowns of Hog Beater Gear Box.							Implemented Area: PW Shopfloor															
Problem/Present Status: By using Tire coupling, vibrations from Hog beater shaft is transferring directly to Gear Box. Due to this there is frequent failure of Gear Box.			Before Improvement: 				Implemented by: C. Prasadhan(Fitter)															
Root Cause Identification: Problem •Frequent Hog Beater Gear Box failure. Why 1 •Gear Box is experiencing vibrations. Why 2 •Hog beater shaft is transferring vibrations to Gear Box. Why 3 •Hog Beater has Tire coupling and it is not absorbing vibrations and oscillations of shaft.			Result/Benefit: (a) Qualitative With reduction in frequent failure of Hog Beaters generation of NC Products has reduced.																			
			After Improvement: 				(b) Quantitative  Breakdown Frequency/ Month Before: 16, After: 3 Cost Saving = 9.76 Lakhs per Year															
Root cause			Tire coupling is not absorbing vibrations and oscillations of the shaft.				Standardization: Preventive Maintenance as per schedule.															
Idea to eliminate root cause			To change all the Tire coupling of Hog beater to Universal coupling.				Horizontal Deployment <table border="1" style="width: 100%; text-align: center;"> <tr> <td>BW</td> <td>CW</td> <td>KW</td> <td>LW</td> <td>PW</td> <td>SW</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>				BW	CW	KW	LW	PW	SW						
BW	CW	KW	LW	PW	SW																	
Counter-measure			Replaced all the Tire couplings of Hog beater with Universal joint coupling, which allows flexibility of the shaft while rotating. By this gear box failure is reduced.																			

Thank You